

Advanced (Habanero)

Q1. What is the slope of the line $y = 2x + 3$?

- (A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q2. A line has a slope of -2 and passes through the point $(1, 3)$. What is its equation?

- (A) $y = -2x + 1$
(B) $y = -2x + 5$
(C) $y = 2x - 1$
(D) $y = 2x + 1$
(E) $y = -2x + 3$

Q3. What is the slope of a line that is perpendicular to $y = 4x - 2$?

- (A) $-$

rac14

(A)

rac14

(A) 4

(B) -4

(C) 0

Q4. The table shows the relationship between x and y . What is the slope?

- (A) 2
(B) 3
(C) 4
(D)

rac12

(D)

rac13

Q5. What does a slope of 0 represent?

- (A) A vertical line
(B) A horizontal line
(C) A line with a positive slope
(D) A line with a negative slope
(E) A line with an undefined slope

Q6. Which equation has an undefined slope?

- (A) $y = 2x + 3$
(B) $y = -3x - 2$
(C) $x = 4$
(D) $y = x - 2$
(E) $y = -x + 1$

Q7. A water tank can hold 1000 gallons of water. If 5 gallons of water are added every minute, what is the rate of change of the amount of water in the tank?

- (A) 5 gallons/minute
(B) 10 gallons/minute
(C) 50 gallons/minute
(D) 100 gallons/minute
(E) 200 gallons/minute

Q8. What is the slope of the line that passes through the points $(2, 3)$ and $(4, 7)$?

- (A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Open Ended Questions (FRQ)

Q9. Find the equation of the line that passes through the point $(1, 2)$ and has a slope of -3 . Show your work and write the equation in slope-intercept form $y = mx + b$.

Q10. A bakery is having a sale on bread. The original price of a loaf of bread is

2.50

, and it is on sale for

1.75

. If the price decreases by

0.25

every week, what is the rate of change of the price of bread? Write your answer as a decimal and show your work.

Chef's Tips

- Read each question carefully before answering.
- Use the space provided to show your work for free response questions.
- Check your work before submitting your test.